

Filippo Ricci

Mathematics — Partial Differential Equations & Calculus of Variations

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📍 Rome, Italy

Research Profile

Master's graduate in Mathematics from the University of Trieste, with a curriculum in Advanced Mathematics and a strong background in partial differential equations, calculus of variations, functional analysis, geometric measure theory, and hyperbolic conservation laws.

My Master's research focuses on the construction of a global weak solution to the shallow-water system in a rotating frame through a variational time-discretization scheme based on the minimization of the mechanical energy of the system.

I have been awarded a PhD scholarship in Mathematics at the Gran Sasso Science Institute (GSSI), where I will begin my doctoral studies in November 2026 under the supervision of Prof. Sara Daneri. My position is funded within the FIS3 project SEASTAR (*SElf-ASsembled paTterns: Analysis and Regularity*).

Education

M.Sc. in Mathematics

University of Trieste

Sep. 2024 – Jul. 2026

Trieste, Italy

- Final grade: **110/110 cum laude**.
- Curriculum: *Advanced Mathematics*.
- Master's thesis: *A Variational Approach to the Shallow-Water Equations via an Energy Minimization Process*.
- Supervisor: **Prof. Stefano Bianchini**, SISSA — International School for Advanced Studies.

Diploma, Percorso Formativo Comune

International School for Advanced Studies

Sep. 2024 – Jul. 2026

Trieste, Italy

B.Sc. in Mathematics

Sapienza University of Rome

Aug. 2021 – Oct. 2024

Rome, Italy

- Final grade: **110/110**.
- Curriculum: *Generale*.
- Bachelor's thesis: *Insiemi di Perimetro Finito (Sets of Finite Perimeter)*.
- Supervisor: **Prof. Annalisa Malusa**, Sapienza University of Rome.

Awards & Scholarships

PhD Scholarship in Mathematics

2026

Gran Sasso Science Institute (GSSI), L'Aquila, Italy

Awarded through competitive public selection. The scholarship is funded within the FIS3 project **SEASTAR – SElf-ASsembled paTterns: Analysis and Regularity**, led by Prof. Sara Daneri.

SISSA Scholarship — Percorso Formativo Comune

2024–2026

SISSA — International School for Advanced Studies / University of Trieste

Awarded a competitive scholarship to attend advanced Master's-level courses within the *Percorso Formativo Comune*, jointly organized with SISSA and the University of Trieste.

Research Experience

Master's Thesis Research

University of Trieste / SISSA

A Variational Approach to the Shallow-Water Equations via an Energy Minimization Process

- Studied the shallow-water system in a rotating frame, with the aim of constructing global weak solutions.
- Developed a variational time-discretization scheme in which, at each time step, the evolution is determined through the minimization of the mechanical energy of the system.
- The variational construction is inspired by the optimal transport approach introduced by Gangbo and Westdick-Engberg for the system of isentropic Euler equations.

- Used tools from partial differential equations, calculus of variations, and optimal transport to construct a global weak solution to the system.

Supervisor: Prof. Stefano Bianchini (SISSA, Trieste).

Selected Graduate Coursework

Functional Analysis
Calculus of Variations
Hyperbolic Conservation Laws

Partial Differential Equations
Geometric Measure Theory
Stochastic Calculus

Teaching Experience

University Tutor

University of Trieste

Oct. 2025 – Dec. 2025

Trieste, Italy

- Provided tutoring sessions for first-year university students.
- Prepared weekly problem sets and exercises.

Private Tutor in Mathematics and Physics

Private teaching

Nov. 2021 – May 2022

Rome, Italy

Provided after-school mathematics and physics tutoring to secondary-school students.

Technical Skills

Programming C++, MATLAB

Office software Microsoft Word, Excel, PowerPoint

Languages

Italian Native

English B2 — listening, reading, writing, spoken production and interaction

Upcoming Position

PhD in Mathematics

Gran Sasso Science Institute (GSSI)

Nov. 2026 –

L'Aquila, Italy

Awarded a PhD scholarship in Mathematics through the competitive public selection procedure. Doctoral studies will begin on **2 November 2026**.

Additional Activities

Youth Education and Volunteering

Rome, Italy

Long-term involvement in educational and recreational activities for children and teenagers, including catechesis, summer camps, activity planning, and logistical organization.

Sports

Football training and participation in official FIGC youth competitions; regular gym and swimming activities.